Reimplementation of Visual SLAM in Dynamic Environments Based on Object Detection and Scene Flow

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Problem Statement

Assumption of Static Environments

- Most Visual Simultaneous Localization and Mapping (V-SLAM) systems assume the environment is static.
- This assumption can lead to system drift or failure when dynamic elements like vehicles and pedestrians are present.

Treatment of Moving Objects

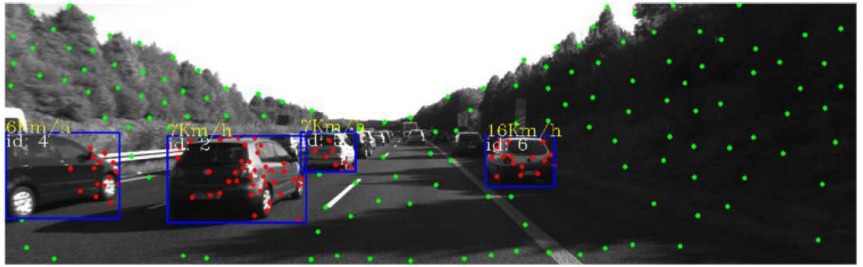
- Dynamic elements are often treated as outliers and excluded, simplifying computation but sacrificing valuable information.
- This results in reduced accuracy of localization.

Challenges in Dynamic Object Identification

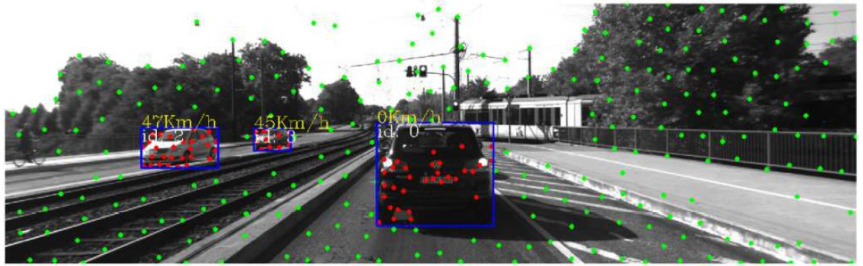
- Distinguishing between truly dynamic objects and static-yet-potentially-dynamic objects (e.g., parked cars) poses a significant challenge.
- This distinction is crucial for robust V-SLAM performance.

Integration Goals

- The goal is to integrate dynamic objects into SLAM state estimation in a theoretically sound manner.

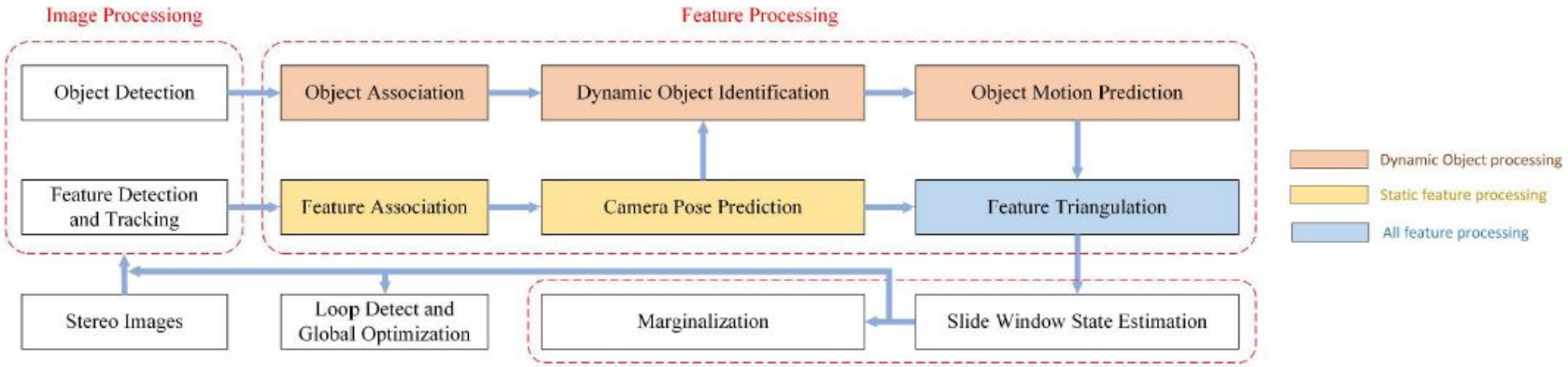


(a)



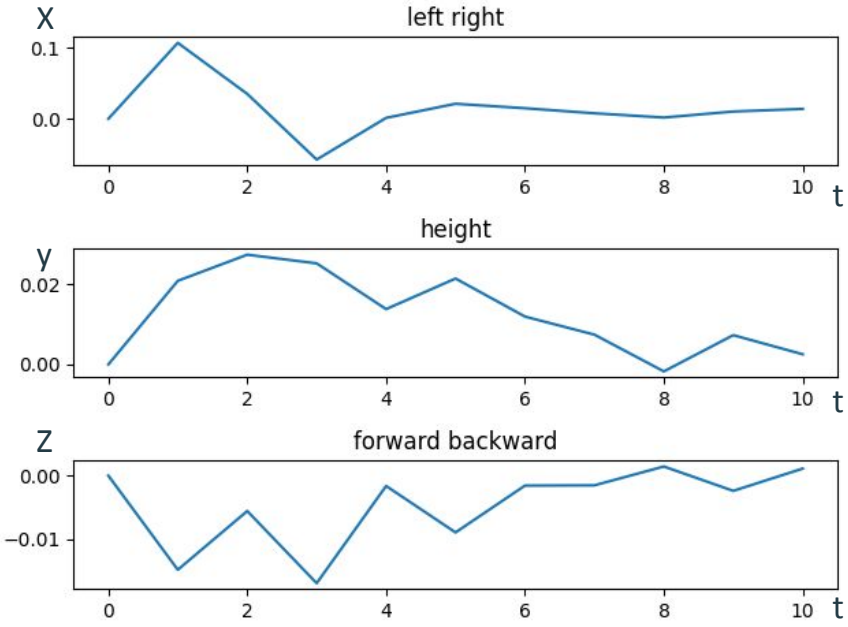
(b)

Suggested method

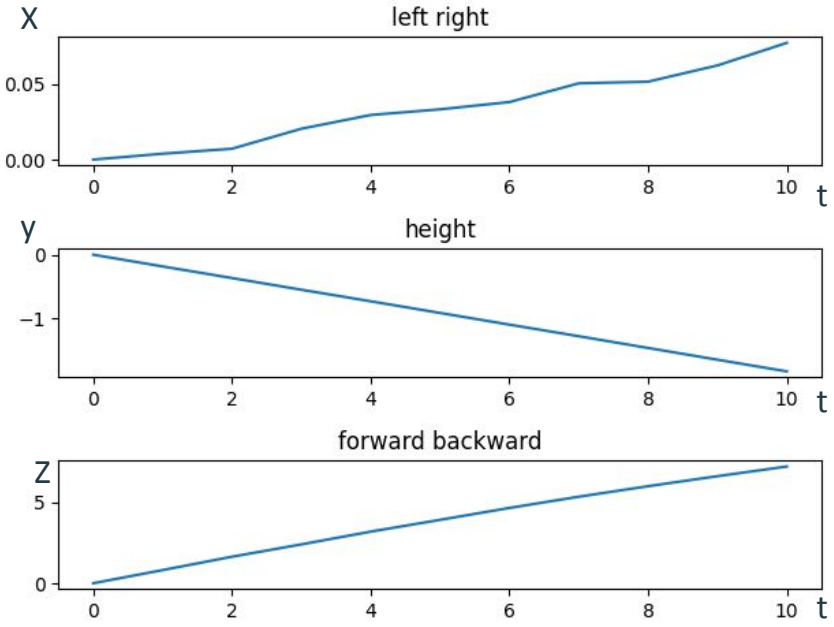


Results - Camera Pose

Our Calculation



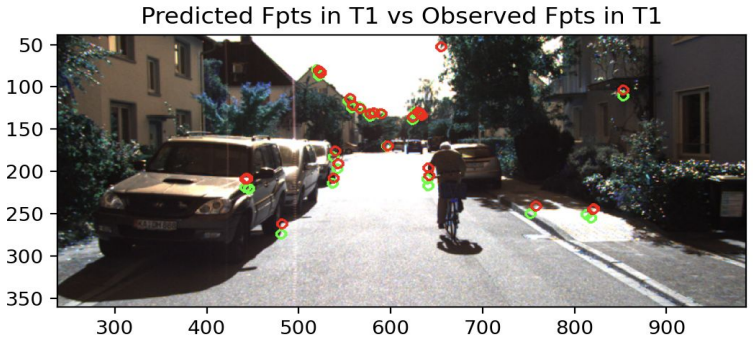
Ground Truth



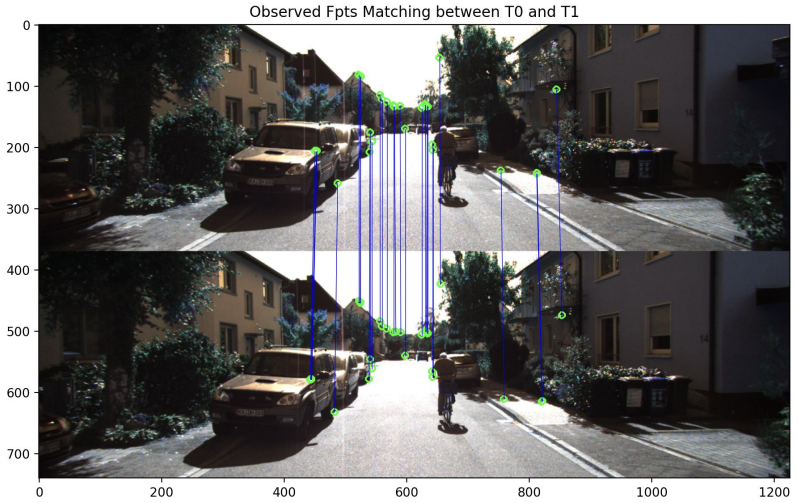
Challenge - Camera Pose

Problem: The Z direction motion is incorrect

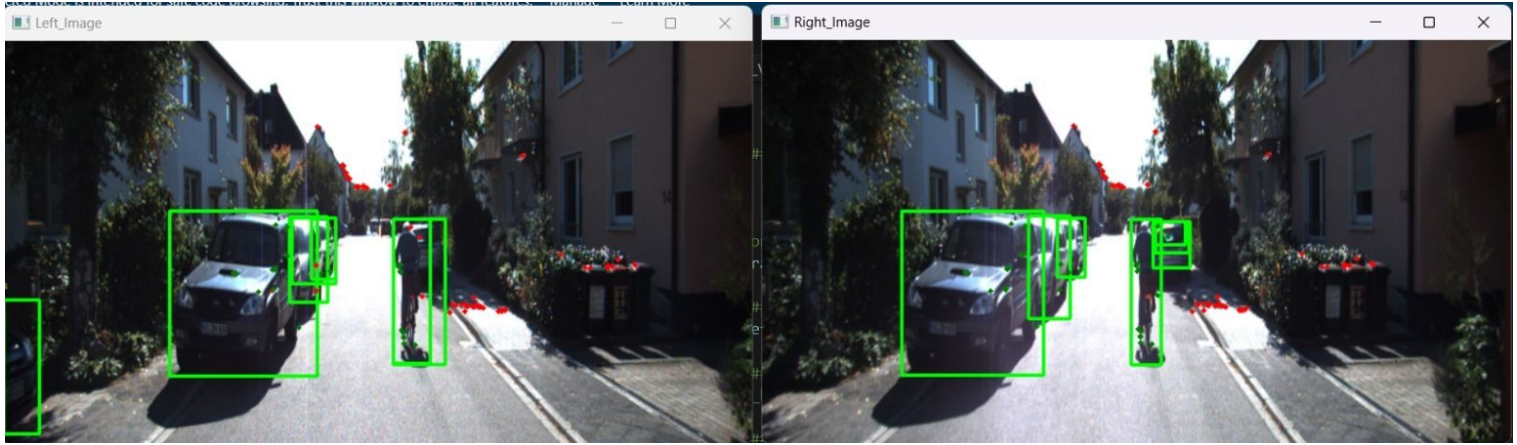
1. Some feature matchings are incorrect
2. The depth we calculated might be incorrect



Green: Predict Red: Observed

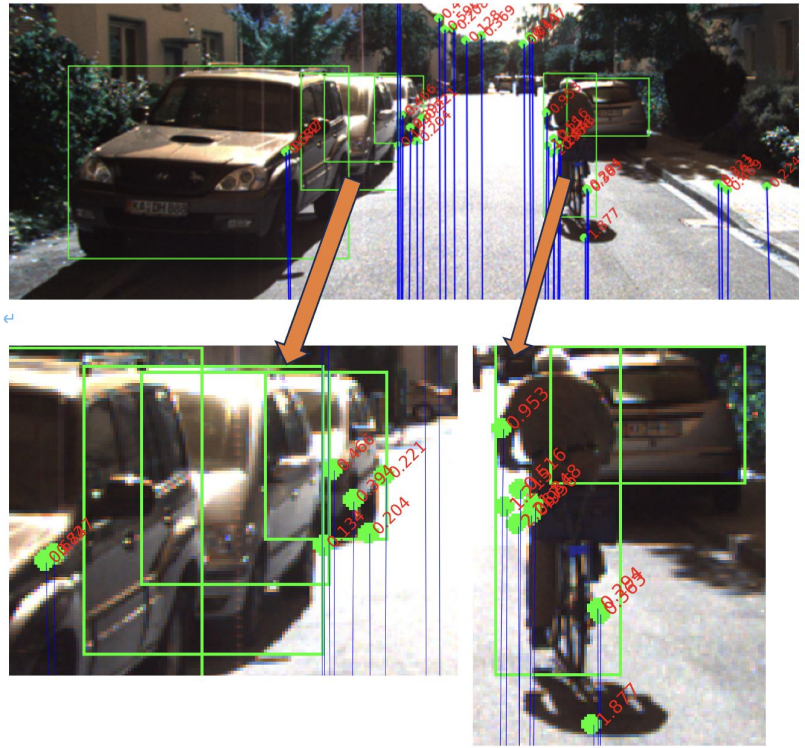


Result - Kmeans



Results - KL





Future Steps

1. Optimize feature matching and Camera Pose Estimation
2. Local Pose Estimation after KL
3. Optimize Kmeans

Thank you !